Wineador controller, Rev Q. Nets shared between sheets use global labels; every net is named.
Generated from tools/revQ/design.py — the netlist report beside this project lists every net and its pins.

Power

Power: 12V input, hold-up, 5V, three 3.3V rails

File: Power.kicad_sch

MCU_USB

ESP32-S3 module, EN and BOOT, supervisor/watchdog, USB-C

File: MCU_USB.kicad_sch

Loads

Loads: interlock, MOSFET outputs, blower stage, plenum fan

File: Loads.kicad_sch

Sensors_RTC

I2C mux, sensor connectors, RTC and backup cell

File: Sensors_RTC.kicad_sch

LCD_SD

LCD connector and level shifter, micro-SD

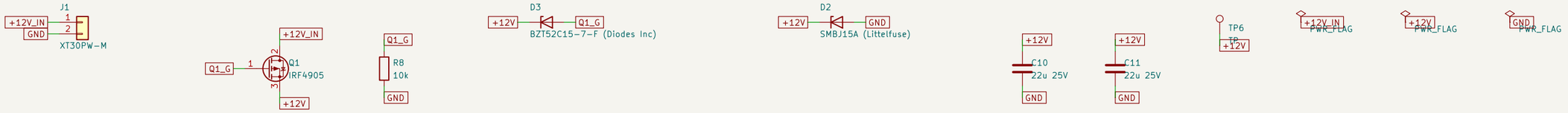
File: LCD_SD.kicad_sch

Root: one sheet per function
Wineador controller
Sheet: /
File: wineador.kicad_sch

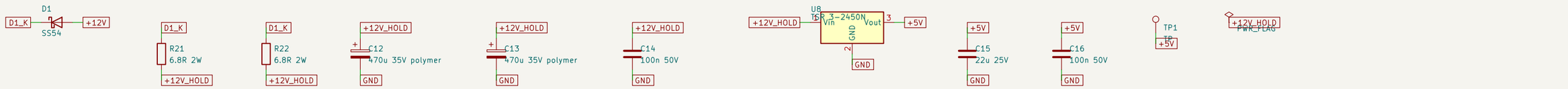
Title: Wineador controller board

Size: A3	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 1/6

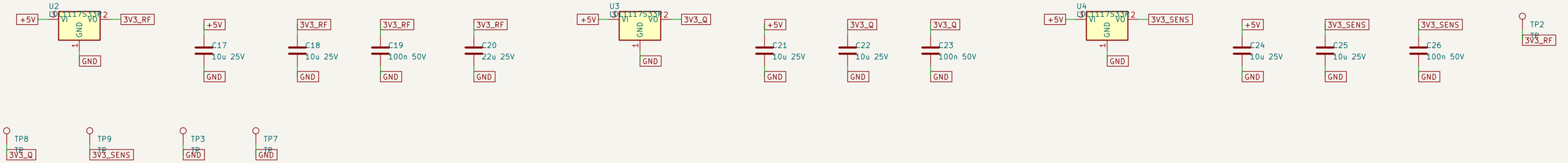
12V input: reverse protection and surge clamp (as Rev P; Q1 through-hole)



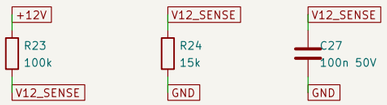
Hold-up branch and 5V regulator. Inrush limiter: 2 x 6.8R 2W in parallel = 3.4R, 4W



3.3V regulators: RF (ESP32 only), Q (SD, mux, RTC, pull-ups), SENS (sensor leads)



12V rail sense to ADC (12V reads 1.57V; 24V clamp level reads 3.2V)



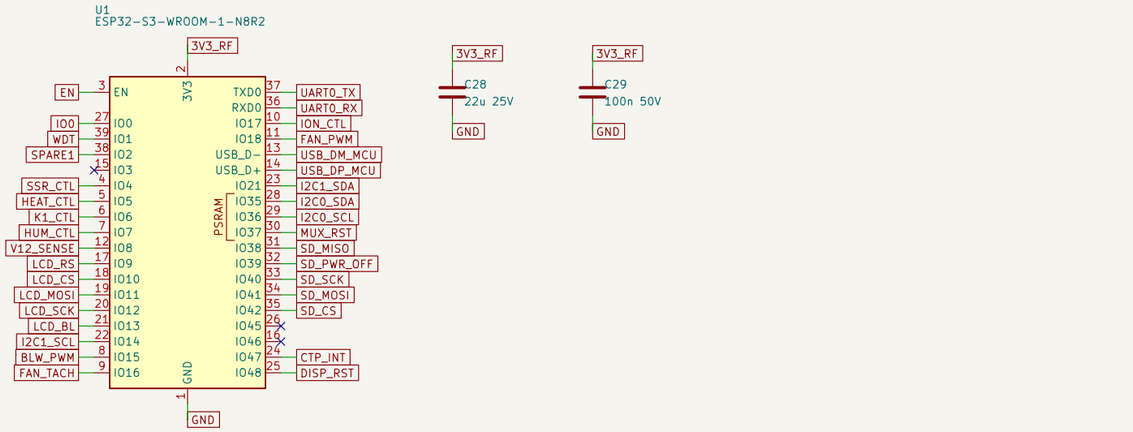
Sheet 2: Power
Wineador controller

Sheet: /Power/
File: Power.kicad_sch

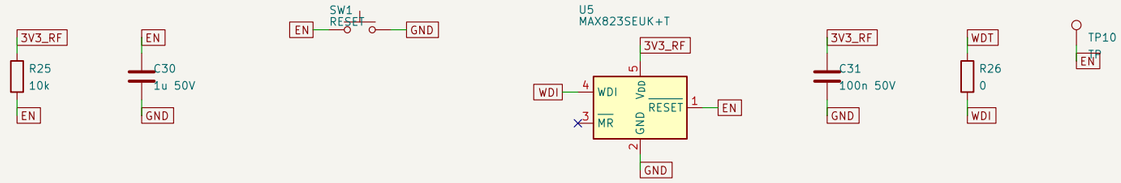
Title: Power: 12V input, hold-up, 5V, three 3.3V rails

Size: A3	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 2/6

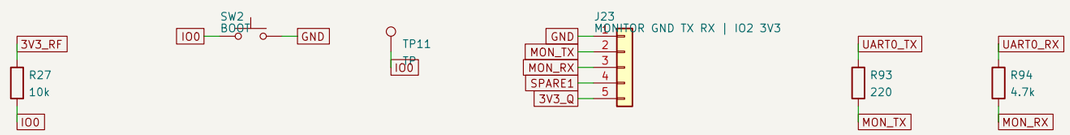
ESP32-S3-WROOM-1-N8R2 (do not substitute N16R8: it takes GPIO35-37)



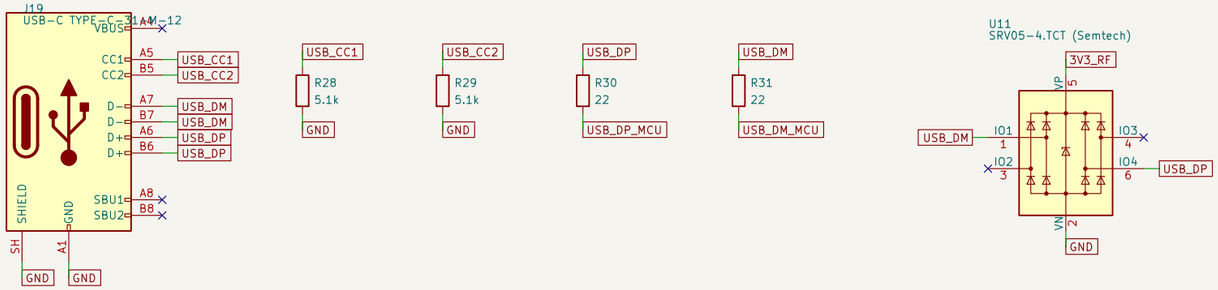
EN net: pull-up, capacitor, reset button, supervisor -- nothing else



BOOT (GPIO0). And J23, the serial monitor header: 1 GND, 2 TX (GPIO43), 3 RX (GPIO44), then 4 GPIO2 spare, 5 3V3_Q

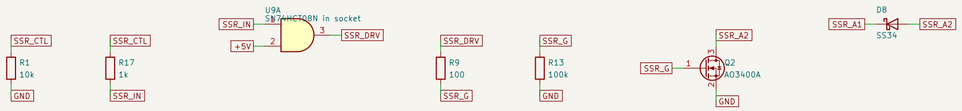


USB-C: data only. VBUS is not connected to anything

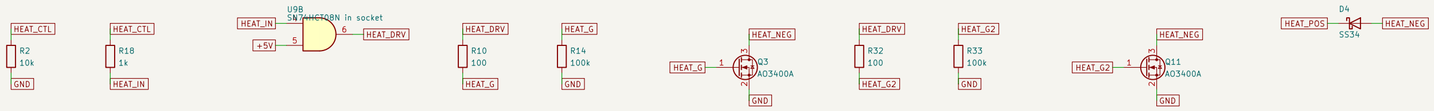


Sheet 3: MCU_USB		
Wineador controller		
Sheet: /MCU_USB/		
File: MCU_USB.kicad_sch		
Title: ESP32-S3 module, EN and BOOT, supervisor/watchdog, USB-C		
Size: A3	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 3/6

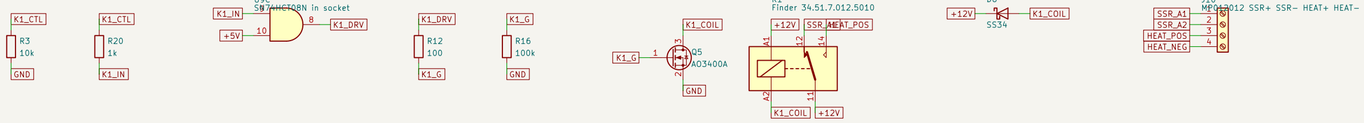
Cooling: SSR control, low side. Fed only when K1 is de-energised



Heater, low side: two AO3400A in parallel, each with its own gate resistor. Fed only when K1 is energised



Interlock relay K1: COM +12V, NC to SSR +A1, NO to heater + (unchanged from Rev P)



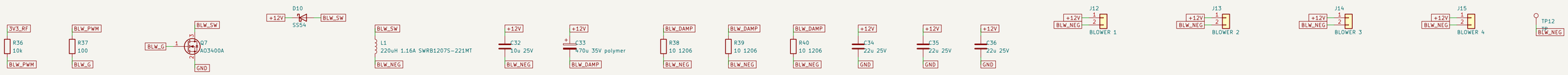
Humidifier fan, low side



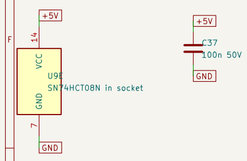
Ion generator, low side, driven straight from GPIO17. Pulled down: off by default



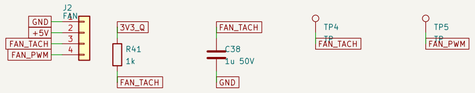
Blowers: smoothed-PWM step-down, low side, driven straight from GPIO15. Pulled UP: undriven = full speed



74HCT08 power (one chip, all four gates used, as Rev P)



Plenum fan J2: 5V, 25kHz PWM, tach (as Rev P)

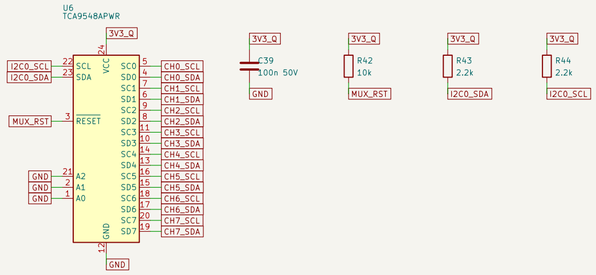


Sheet 4: Loads
Wineador controller
Sheet: /Loads/
File: Loads.kicad_sch

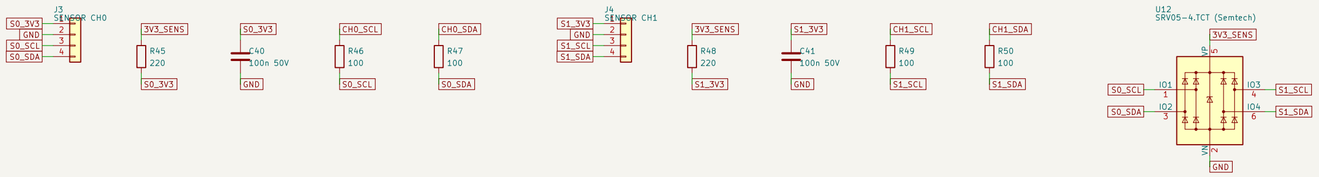
Title: Loads: interlock, MOSFET outputs, blower stage, plenum fan

Size: A2	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 4/6

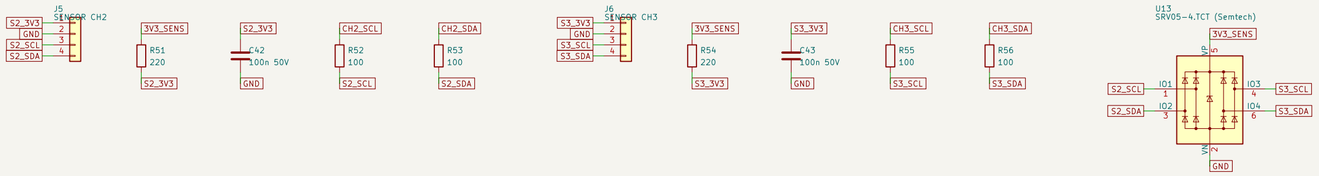
I2C mux TCA9548A at 0x70 on I2C0 (GPIO35 SDA, GPIO36 SCL). No channel pull-ups



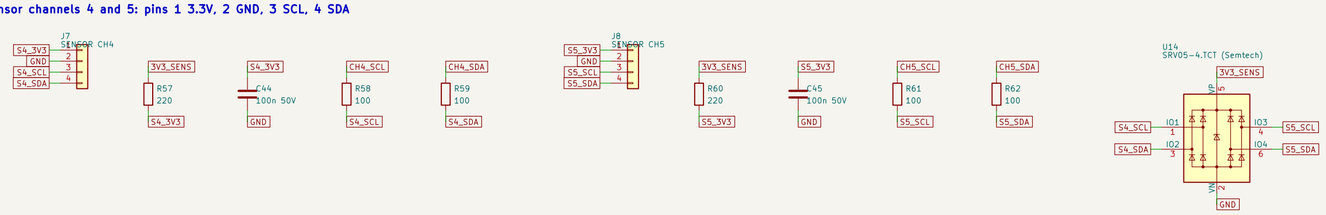
Sensor channels 0 and 1: pins 1 3.3V, 2 GND, 3 SCL, 4 SDA



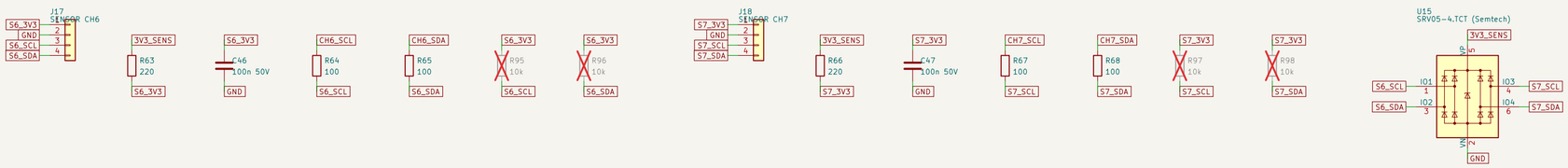
Sensor channels 2 and 3: pins 1 3.3V, 2 GND, 3 SCL, 4 SDA



Sensor channels 4 and 5: pins 1 3.3V, 2 GND, 3 SCL, 4 SDA



Sensor channels 6 and 7 (spare connectors): pins 1 3.3V, 2 GND, 3 SCL, 4 SDA



RTC DS3231SN on I2C1 (GPIO21 SDA, GPIO14 SCL), ML-2020 cell charged from 3V3_Q

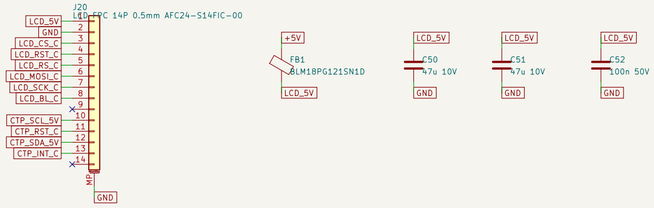


Sheet 5: Sensors_RTC
Wineador controller
Sheet: /Sensors_RTC/
File: Sensors_RTC.kicad_sch

Title: I2C mux, sensor connectors, RTC and backup cell

Size: A2	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 5/6

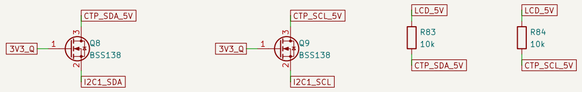
LCD MSP4031 on SPI2: FPC socket (lower contacts: ribbon contacts face the board)



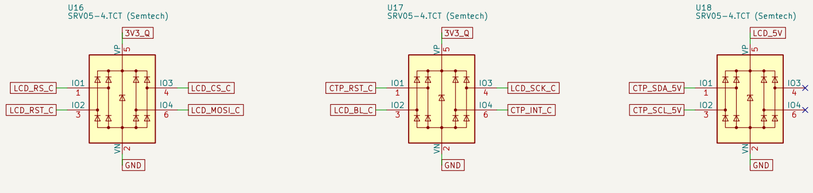
LCD signals: series resistors at the source; one GPIO resets both LCD and touch



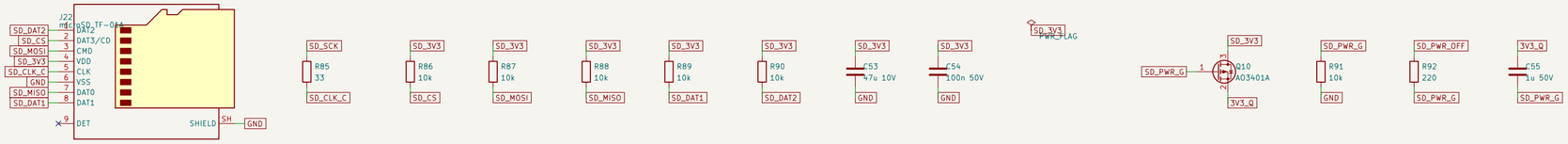
Touch I2C level shifter: the LCD pulls its side to 5V; I2C1 stays at 3.3V. Extra 10k pull-ups to 5V on the cable side



ESD arrays on the LCD cable lines



Micro-SD on SPI3 (GPIO38 MISO, 40 SCK, 41 MOSI, 42 CS), power switched by GPIO39 (undriven = on)



Sheet 6: LCD_SD
Wineador controller
Sheet: /LCD_SD/
File: LCD_SD.kicad_sch

Title: LCD connector and level shifter, micro-SD

Size: A2	Date: 2026-11-11	**Rev: Q**
KiCad E.D.A. 10.0.7		Id: 6/6